

Basic Principles of Medicine 2

Module : DM

DLA:

Introduction to amino acid metabolism

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Recommended reading

- Lippincott's Biochemistry 9th Edition
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
 - Chapter 19: Amino acids: Nitrogen disposal
 - Sections: Overall nitrogen metabolism; Nitrogen removal from amino acids
 - Chapter 20: Amino acid degradation and synthesis
 - Sections: Glucogenic and ketogenic amino acids; Amino acid carbon-skeleton metabolism
- [19. Amino Acids: Nitrogen Disposal | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- [Amino Acids: Degradation and Synthesis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

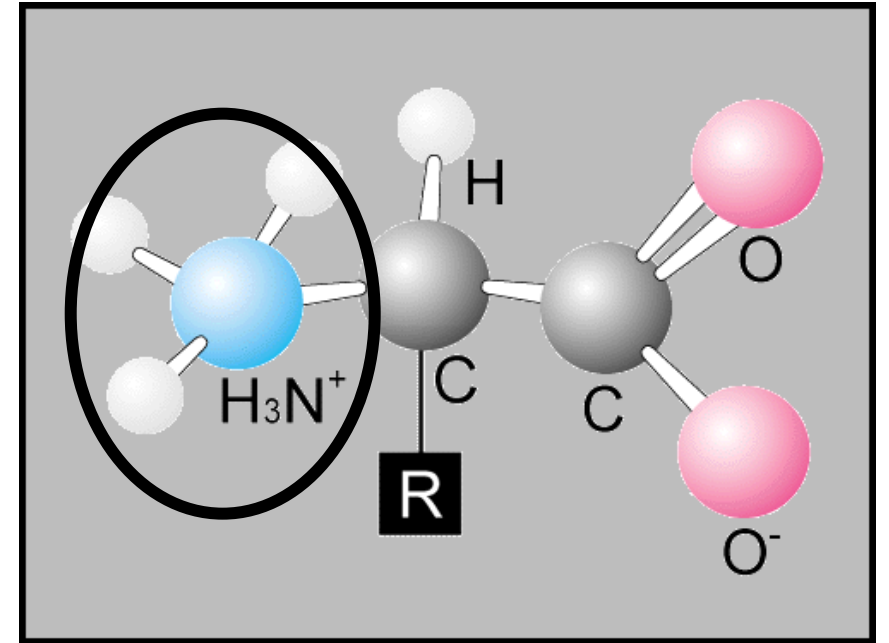
Objectives: DLA: Introduction to Nitrogen Metabolism and Fates of Amino Acids

- Before DM Lecture 23

SOM. MK.I.BPM2.3.DM.2.BCHM.1216	Analyze how amino acids may enter intermediary metabolism and describe 'amino acid pool'
SOM. MK.I.BPM2.3.DM.2.BCHM.1217	Discuss the reactions of alanine amino transferase, aspartate aminotransferase, glutaminase and glutamate dehydrogenase and analyze the central role of these amino acids in metabolism
SOM. MK.I.BPM2.3.DM.2.BCHM.1218	Define and enumerate the essential and nonessential amino acids in humans
SOM. MK.I.BPM2.3.DM.2.BCHM.1220	Distinguish the fates of C- skeletons of amino acids: Glucogenic amino acids, Ketogenic amino acids, Glucogenic & Ketogenic amino acids with examples.
SOM. MK.I.BPM2.3.DM.2.BCHM.1221	Review the TCA cycle intermediates formed from glutamate, glutamine, aspartate, and alanine.

Amino acid/Nitrogen Metabolism

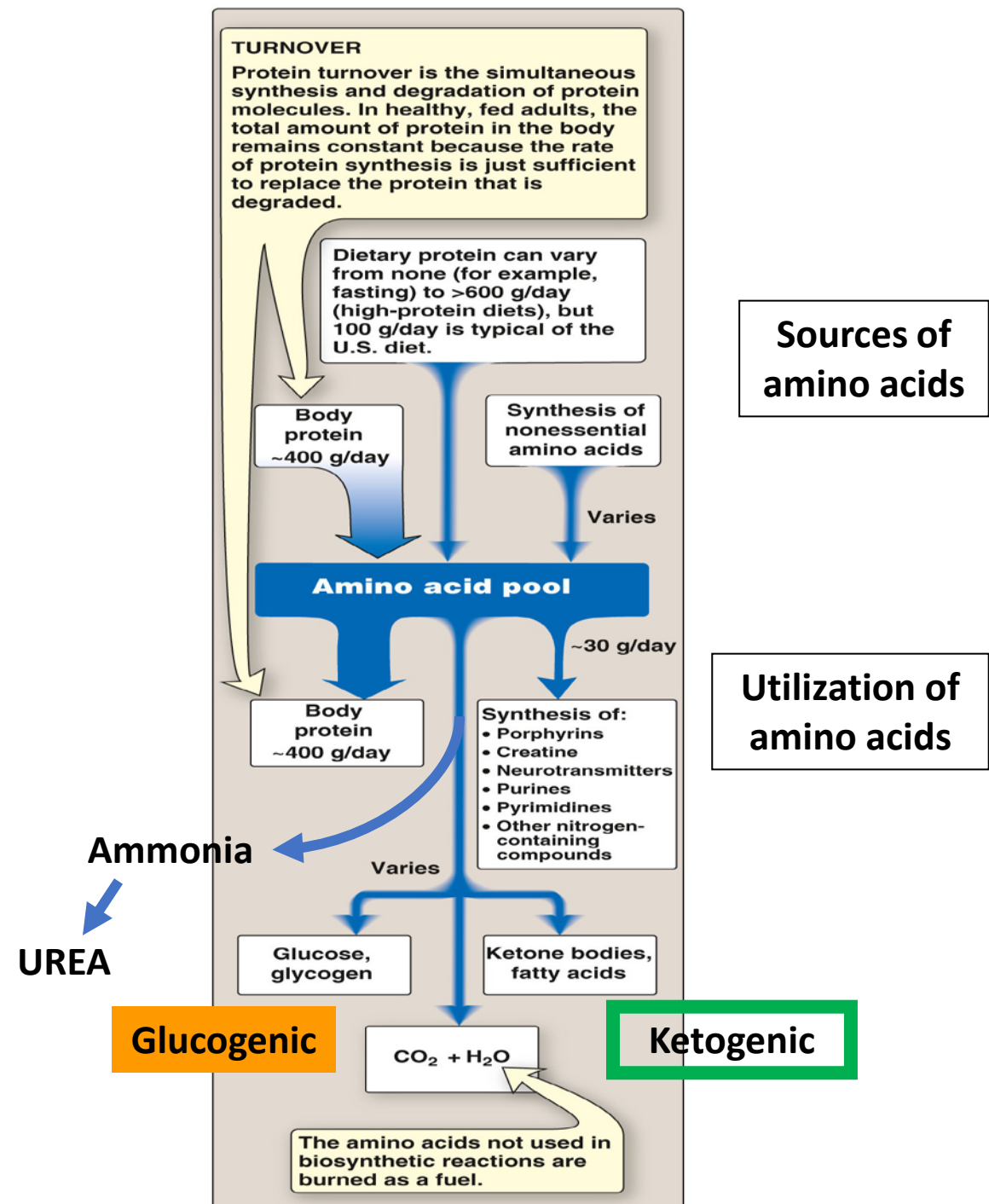
- Tissue protein compartments
 - Somatic protein
 - Visceral protein
- Nitrogen obtained from dietary sources:
 - Proteins (amino acids)
- Needed for growth (protein synthesis)
- Synthesis of amino acid derivatives
 - Creatine, Purines, Heme, Neurotransmitters, and Pyrimidines, many others



There is no storage form of nitrogen in humans

Amino acid pool

- Sources of amino acids
 - Dietary protein
 - Breakdown of tissue protein
 - Synthesis of non-essential amino acids
- Utilization of amino acids
 - Tissue protein synthesis
 - Synthesis of N-containing compounds
 - Catabolism to glucogenic/ ketogenic intermediates
- Protein turnover and Nitrogen balance
 - Positive and negative nitrogen balance



Classification of Amino Acids

- Dietary requirement: Essential vs Non-essential Amino Acids
 - Essential: Cannot be synthesized by humans
 - Non-essential: Can be synthesized by humans
- Metabolic fate: Glucogenic vs Ketogenic Amino Acids
 - Glucogenic: Catabolism yields pyruvate or one of the intermediates of citric acid cycle (**Substrates for gluconeogenesis**)
 - Ketogenic: Yields acetoacetate (ketone body), acetyl CoA or acetoacetyl CoA (**Not Substrates for gluconeogenesis**)

Nonessential

Essential

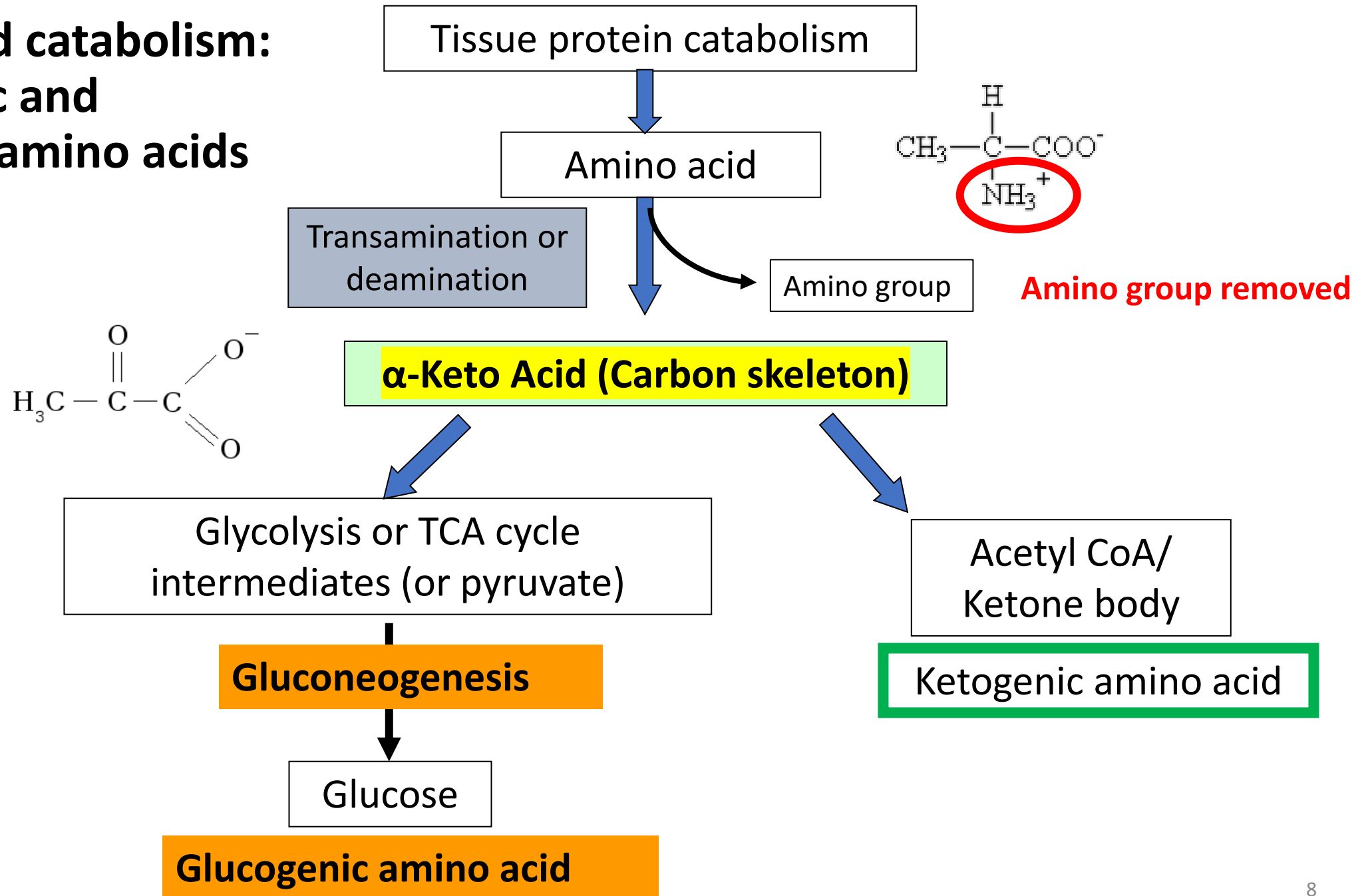
Glucogenic	Glucogenic and Ketogenic	Ketogenic
Alanine Arginine Asparagine Aspartate Cysteine Glutamate Glutamine Glycine Proline Serine	Tyrosine	
Histidine Methionine Threonine Valine	Isoleucine Phenyl- alanine Tryptophan	Leucine Lysine

Amino acids not synthesized in humans: Dietary essential

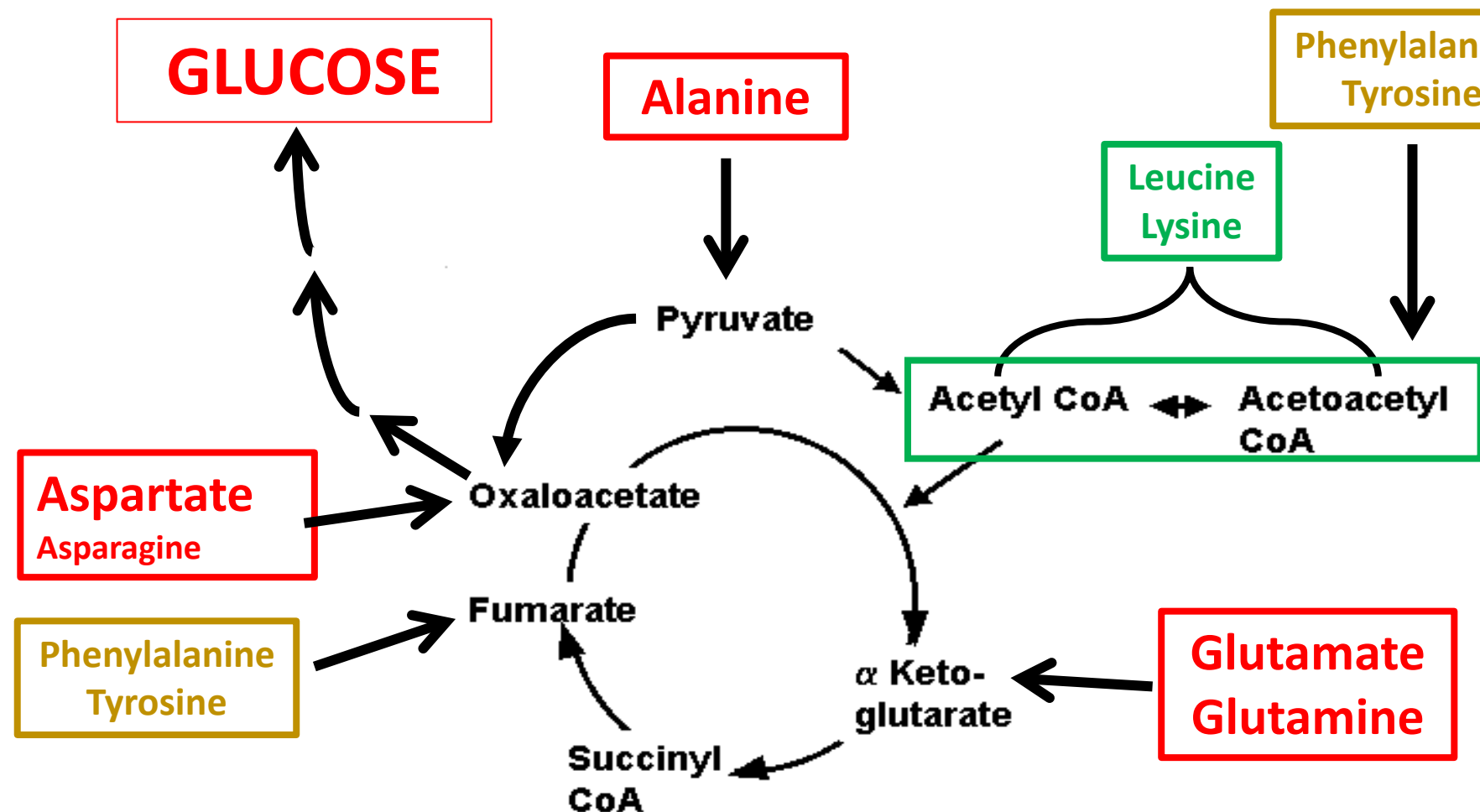
- Cys and Tyr conditionally essential:
 - Low Met in diet, Cys becomes essential
 - Low Phe in diet, Tyr becomes essential
- Essential amino acids: PVT TIMHALL
 - Except Arginine (non-essential)

PVT TIMHALL

Amino acid catabolism: Glucogenic and Ketogenic amino acids



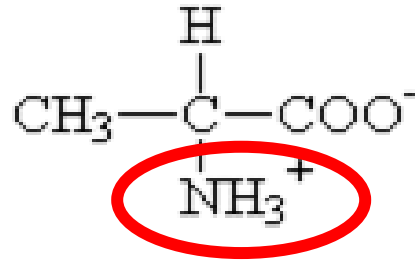
Metabolic fate of C-skeleton: **Glucogenic**, **Ketogenic** or **both**



	Glucogenic	Glucogenic and Ketogenic	Ketogenic
Nonessential	Alanine Arginine	Tyrosine	
	Asparagine Aspartate		
	Cysteine		
	Glutamate Glutamine		
	Glycine		
	Proline Serine		
Essential	Histidine Methionine Threonine Valine	Isoleucine Phenylalanine Tryptophan	Leucine Lysine

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Alanine Metabolism



Alanine

α-ketoglutarate

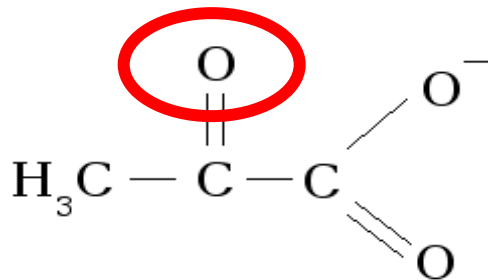
Transamination

Glutamate

Alanine aminotransferase
Alanine transaminase (ALT)

1

Pyridoxal phosphate (PLP; Vit B6)



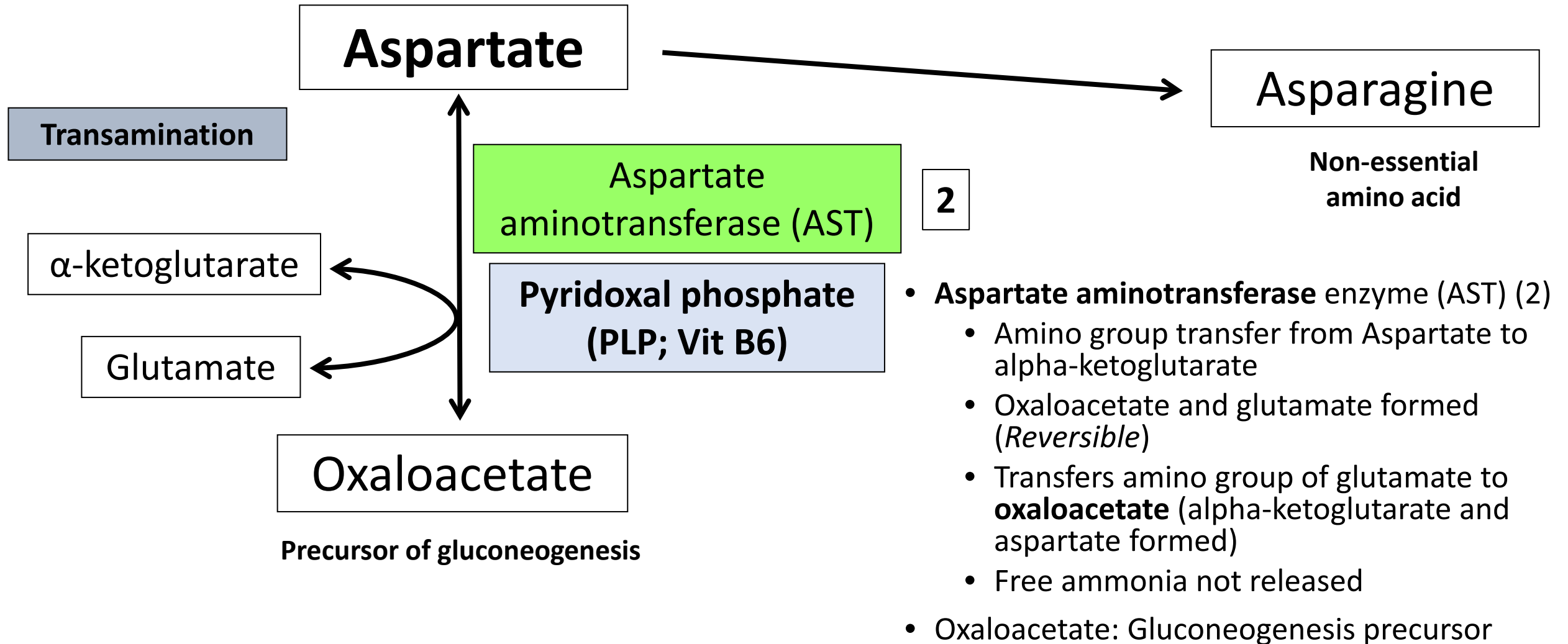
Pyruvate

Amino group of alanine transferred to α-ketoglutarate, to form glutamate. Reversible reaction

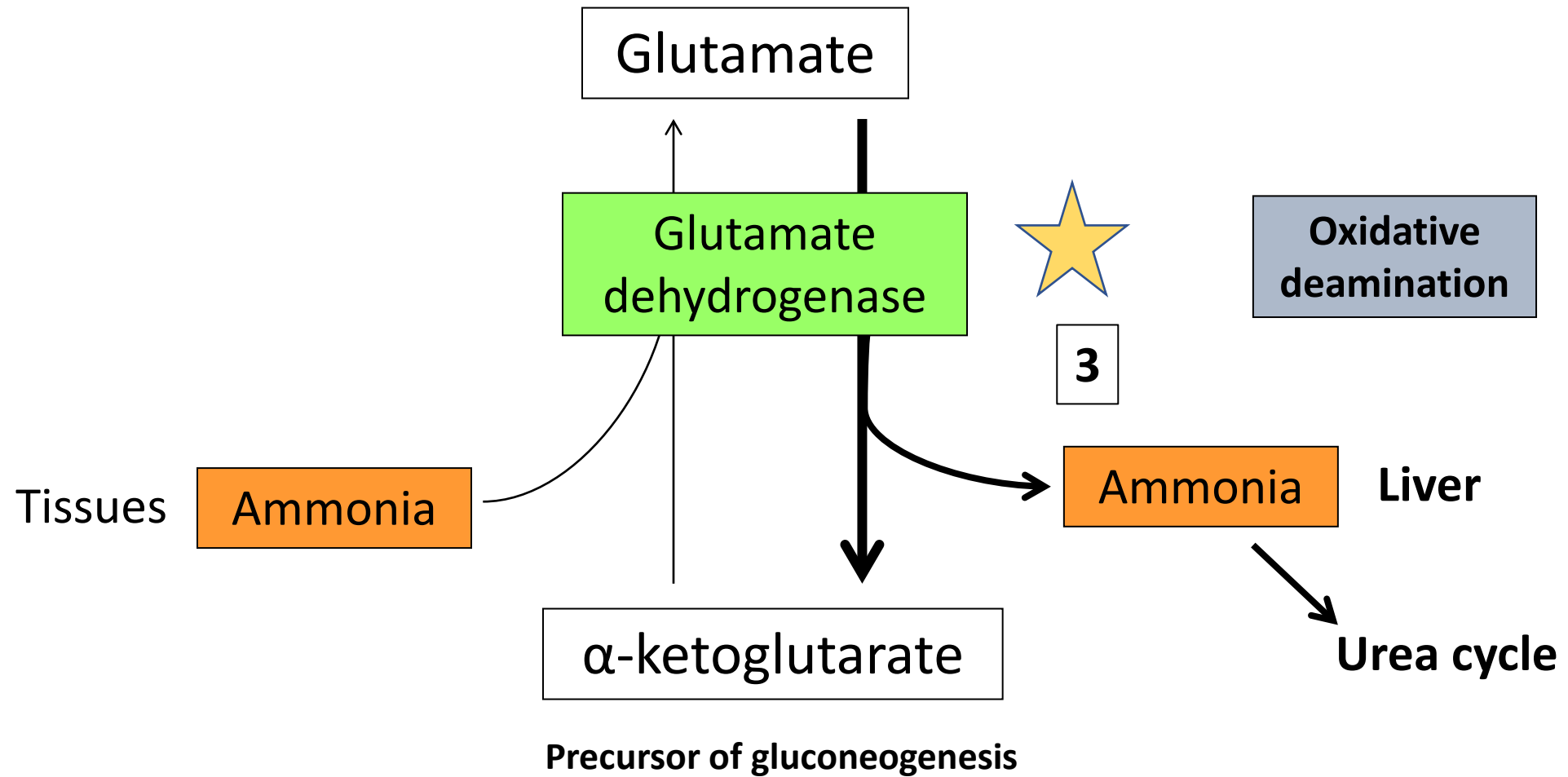
Alanine Metabolism

- **Alanine transaminase** enzyme (ALT) (1)
 - Amino group transferred from alanine to alpha-ketoglutarate
 - **Pyruvate** and glutamate formed (*Reversible*)
 - Transfers amino group of glutamate to **pyruvate** (alpha-ketoglutarate and alanine formed)
 - Free ammonia NOT released
- Alanine : Transport form of amino acid (and nitrogen) from **muscle** (important during starvation)
- Alanine (**pyruvate**): precursor for gluconeogenesis during starvation
- Glucose-alanine cycle involves the **muscle** (where alanine is formed) & **liver** (site of gluconeogenesis)

Aspartate Metabolism

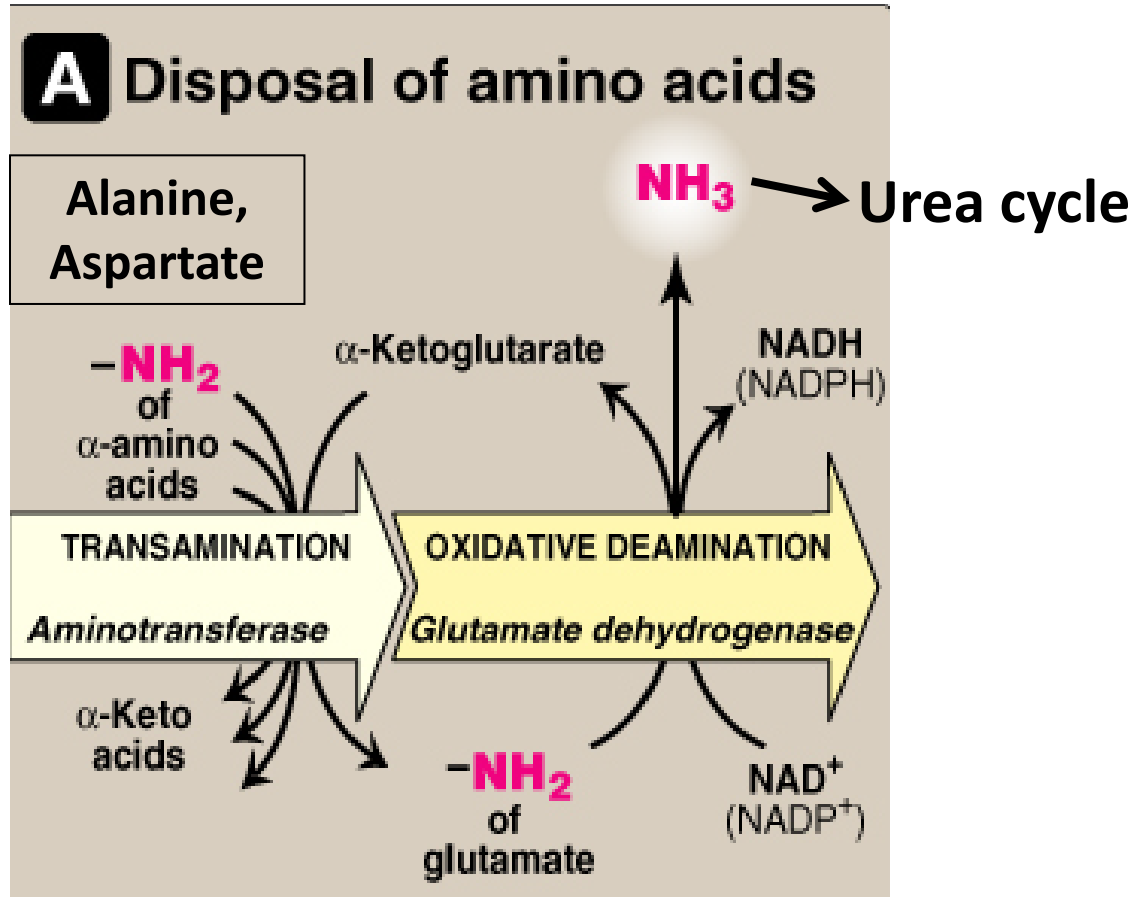


Glutamate Metabolism: Glutamate dehydrogenase enzyme



Oxidative deamination by *glutamate dehydrogenase* is reversible

Liver



1

2

3

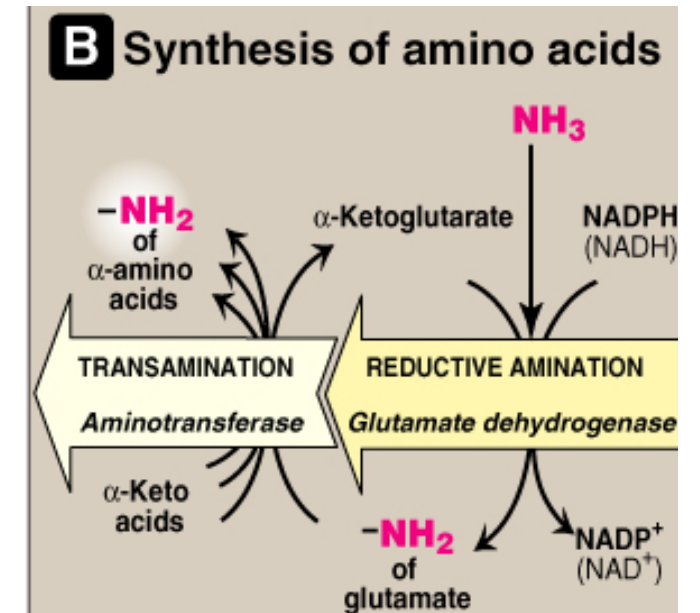
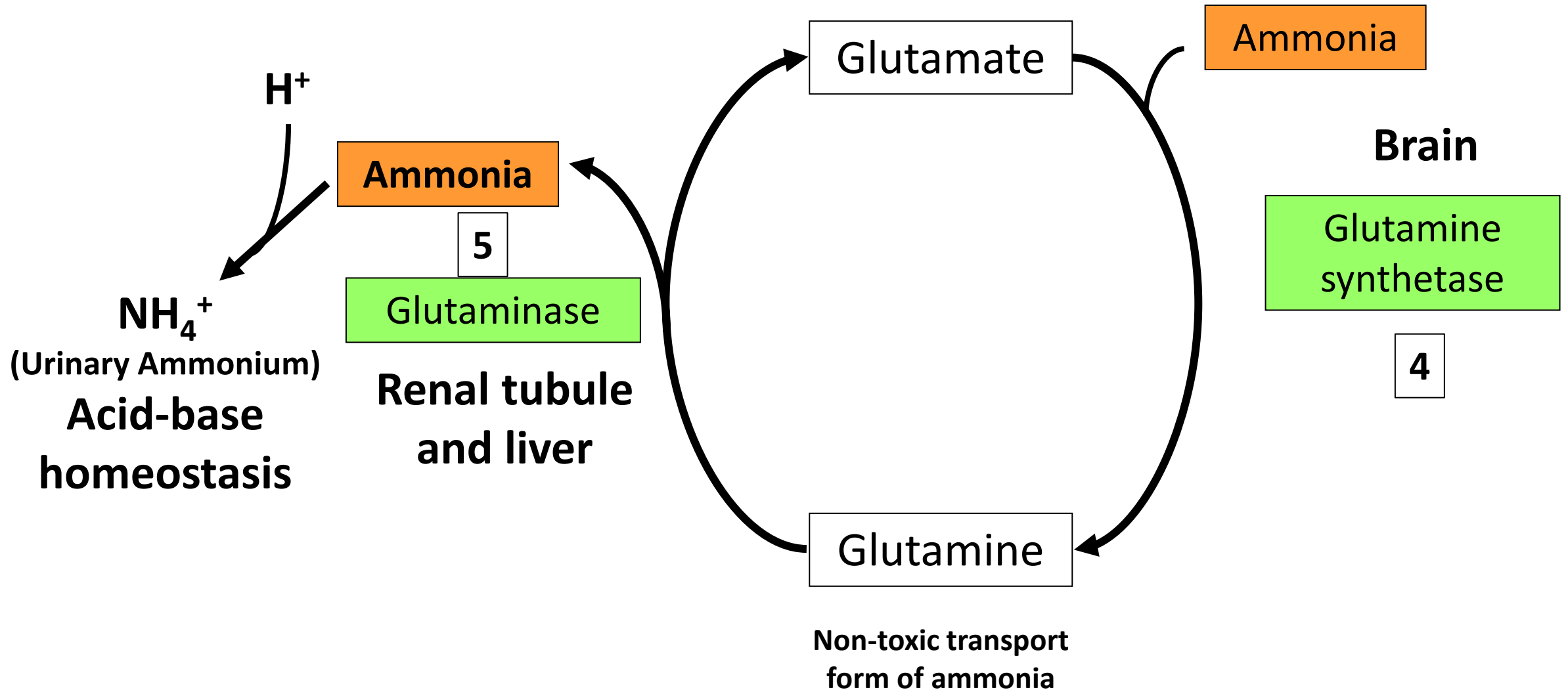


Figure 19.12
Lippincott

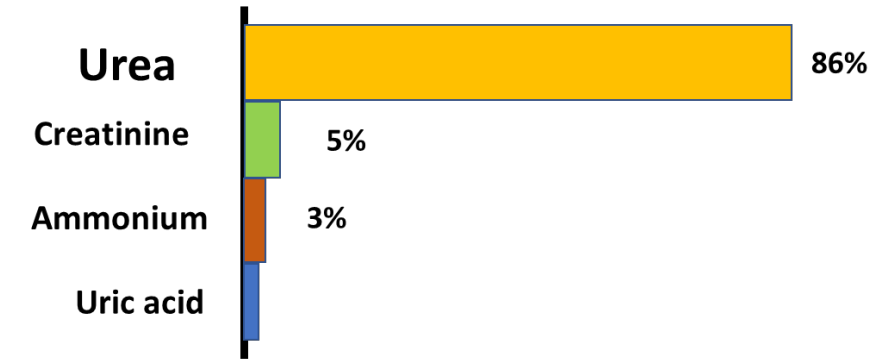
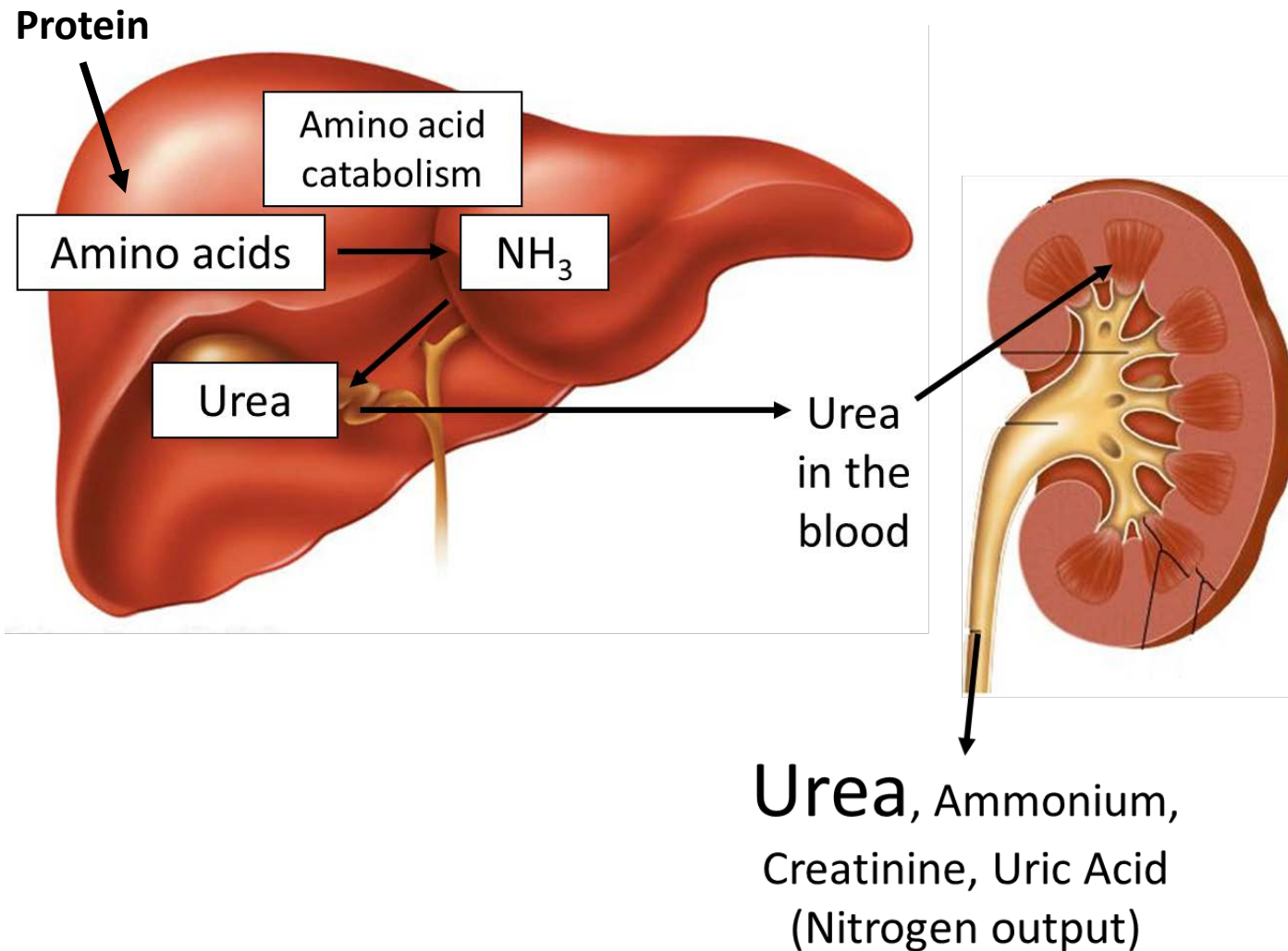
Glutamate and Glutamine



Glutamate and Glutamine Metabolism

- **Glutamate** and glutamine: Sources of **ammonia** for the urea cycle
- **Glutamate dehydrogenase (3):**
 - **Oxidative deamination** reaction (requires NAD^+)
 - Source of **ammonia** for the urea cycle
 - **Reversible** and used for glutamate synthesis when tissue ammonia levels are high
- Glutamate and α -ketoglutarate are essential for **transaminase** reactions
- **Glutamine Synthetase (4):** Scavenges NH_3 in **Brain**
- **Glutamine:** Non-toxic **transport form of ammonia** from peripheral tissues (Brain)
- Renal tubular cells rich in **Glutaminase (5):** Provide ammonia for **acid-base homeostasis**
 - Secreted H^+ bind to ammonia to form NH_4^+ (Ammonium) lost in urine

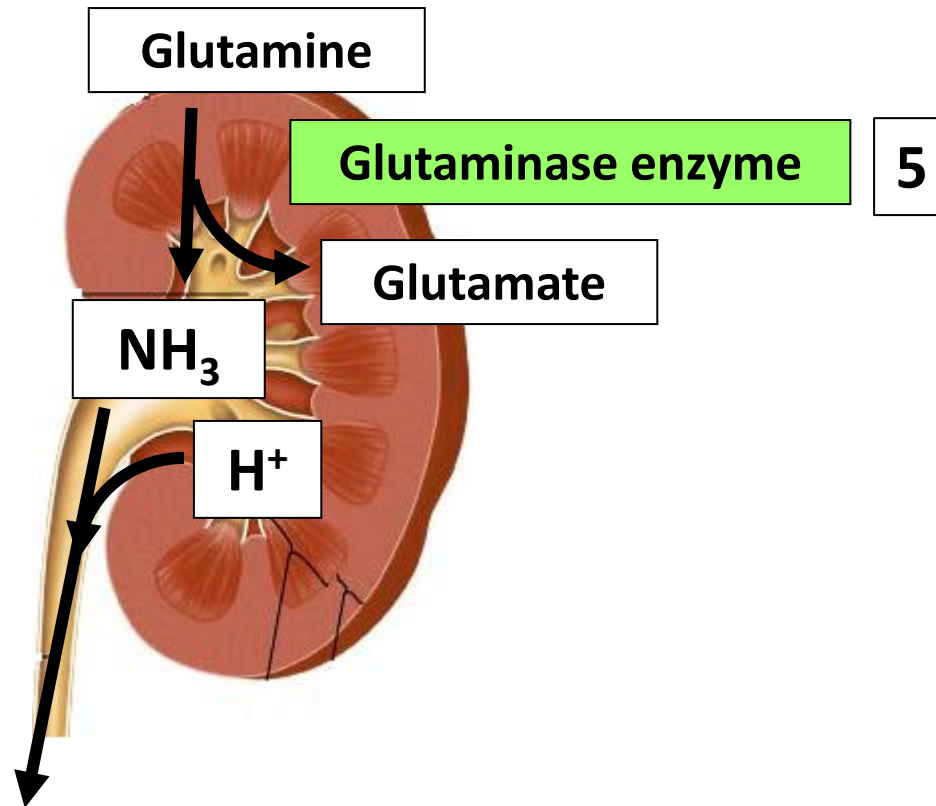
Role of the Liver and Kidney in Nitrogen Metabolism



Non-protein nitrogenous substances excreted by kidney:

- **Urea (Major Nitrogen compound excreted in urine; End-product of amino acid metabolism)**
- Uric acid (end-product of purine degradation)
- Creatinine (end-product of muscle creatine degradation)
- Ammonium (For H⁺ excretion)

Ammonium excretion: Acid-base balance



- Glutaminase (5) releases NH_3
- NH_3 binds to H^+ secreted into tubule $\rightarrow \text{NH}_4^+$
- Chronic acidosis $\rightarrow$ Increased NH_4^+ in urine

NH_4^+ excreted in urine
(Binds to secreted H^+ ion/
Facilitates acid excretion)

Role of the Liver in Nitrogen Metabolism

- Amino acids release their amino groups as ammonia
 - Carbon skeleton is released (For gluconeogenesis or Oxidized in TCA cycle)
- NH_3 detoxified via the Urea cycle in liver (NH_3 is TOXIC)
- **Urea** end-product of **amino acid** metabolism

Role of the Kidney in Nitrogen Metabolism

- **Urea** formed in liver transported to kidneys, and excreted in urine
- Kidneys also excrete H^+ as ammonium ions
 - Regulation of acid base balance
 - Ammonia source is glutamine [via glutaminase (5)enzyme]

Summary:

- Classify amino acids based on:
 - Metabolic fate: Glucogenic/ ketogenic or both
 - Dietary requirement: Essential and non-essential
- Concept of amino acid pool
- Central role of alanine, aspartate and glutamate/glutamine in amino acid metabolism
 - Enzymes utilized, C-skeletons formed, role in metabolism
- Role of liver and kidney in Amino acid/Nitrogen metabolism



Thank you!!

Please email shupadhya@sgu.edu if you have any additional questions

